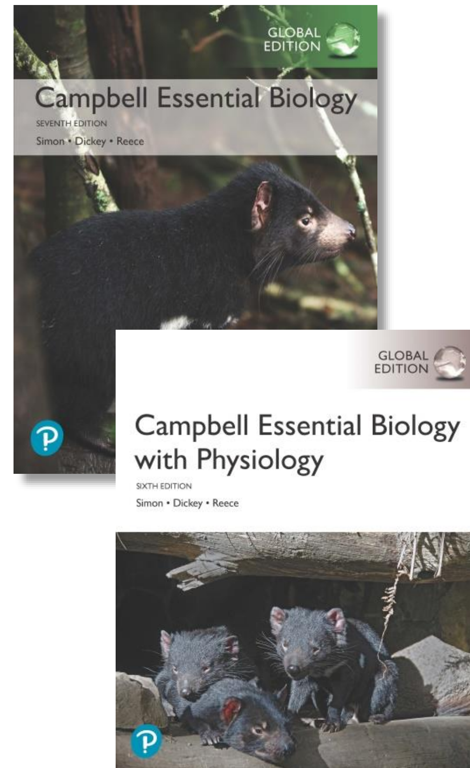
Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 23
Circulation and Respiration



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Pearson

Why is Carbon Monoxide Deadly? It can Hijack the Hemoglobin in your Blood, Displacing the Oxygen you Need to Live

Pearson

Biology and Society: Avoiding "The Wall"

- World-class athletes have world-class circulatory and respiratory systems.
  - To perform work over an extended period, muscle cells require a steady supply of oxygen (O_2) and must continuously eliminate carbon dioxide (CO_2).
  - Stamina of most athletes is limited by the ability of their heart and lungs to deliver the required amount of O_2 to muscle cells. Without enough O_2, athletes will hit "the wall", a sudden loss of energy to the extent that they cannot continue to perform.
  - Properly functioning circulatory and respiratory systems are essential to all of us.

The Demand for Stamina

Unifying Concepts of Animal Circulation

- All life depends on the ability to transform energy and matter. To do this, every organism must exchange materials and energy with its environment.
  - In simple animals, nearly all the cells are in direct contact with the outside world. Thus, each body cell in these organisms can easily exchange materials with the environment by diffusion.
  - Most animals are too large or complex for exchange to occur by diffusion alone. Instead, a **circulatory system** facilitates the exchange of materials.

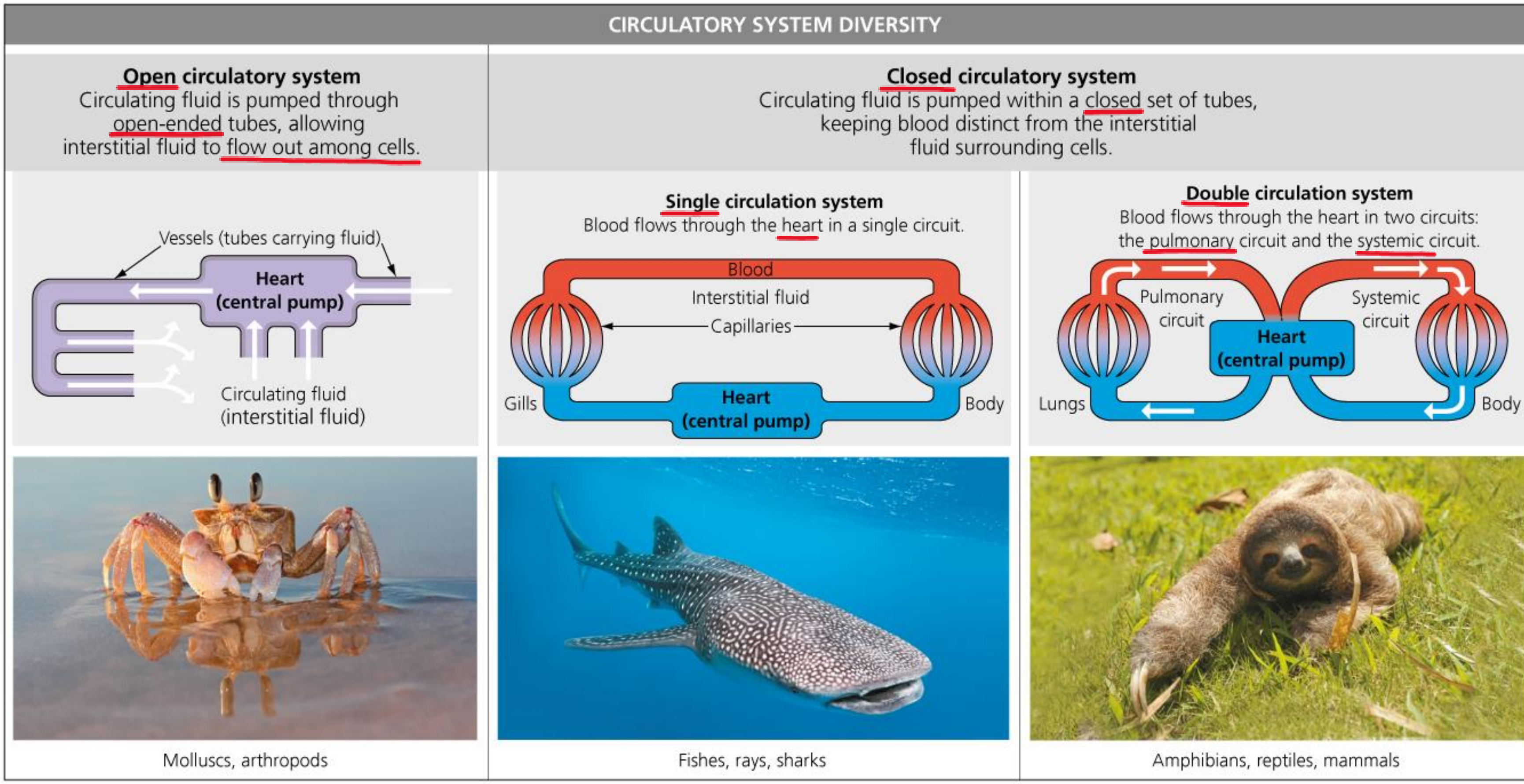
Unifying Concepts of Animal Circulation
(2 of 7)

- All but the simplest animals have a circulatory system with three main components:
    1. a central pump,
    2. a vascular system (a set of tubes), and
    3. a circulating fluid.

Unifying Concepts of Animal Circulation
(3 of 7)

- Two main types of circulatory systems have evolved in animals.
  1. Many invertebrates, including most molluscs and all arthropods, have an **open circulatory system** in which fluid is pumped through open-ended tubes and flows out among cells.
  2. Nearly all other animals have a **closed circulatory system** in which blood is pumped within a set of closed tubes and is distinct from the interstitial fluid.

The Diversity of Circulatory Systems (1 of 2)



Unifying Concepts of Animal Circulation
(4 of 7)

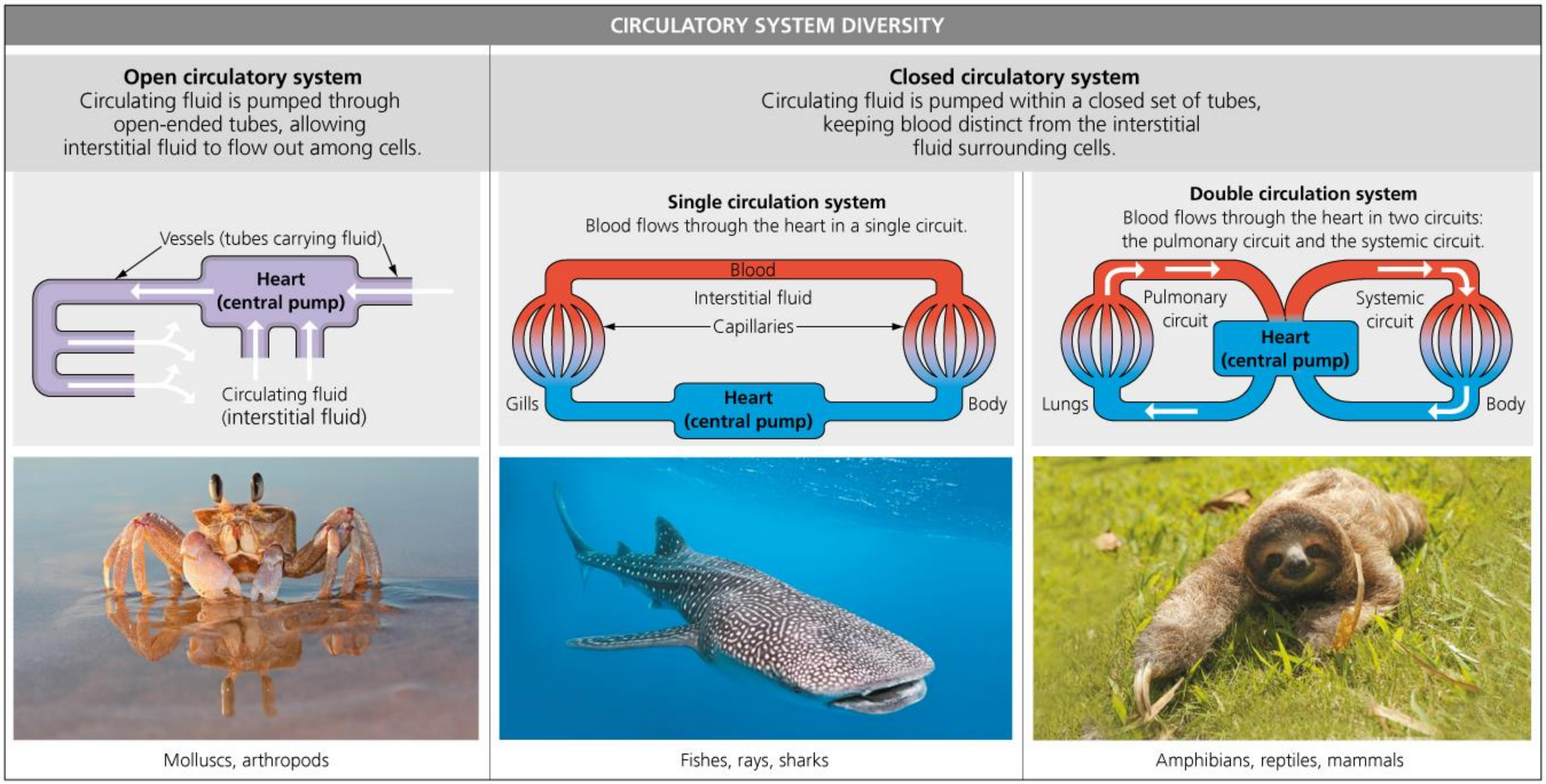
- The closed circulatory system of humans and other vertebrates is called a **cardiovascular system**. It consists of the heart, blood, and three types of blood vessels (arteries, capillaries, and veins).
  1. **Arteries** carry blood away from the heart and branch into smaller arterioles as they near organs.
  2. **Capillaries**, with thin walls, allow exchange between blood and interstitial fluid.
  3. Venules collect blood from the capillaries and converge to form **veins**, which return blood back to the heart, completing the circuit.

Arterioles: 세동맥
Venules: 세정맥

Unifying Concepts of Animal Circulation

- Vertebrates have two distinct types of cardiovascular systems.
  1. In the **single circulation system**, found in bony fishes, rays, and sharks, blood flows only once through the heart by way of a single loop/circuit.
  2. In the **double circulation system**, found in amphibians, reptiles (including birds), and mammals, blood flows twice through the heart,
     - once between the lungs and the heart in the **pulmonary circuit** and
     - a second time between the heart and the rest of the body in the **systemic circuit**.

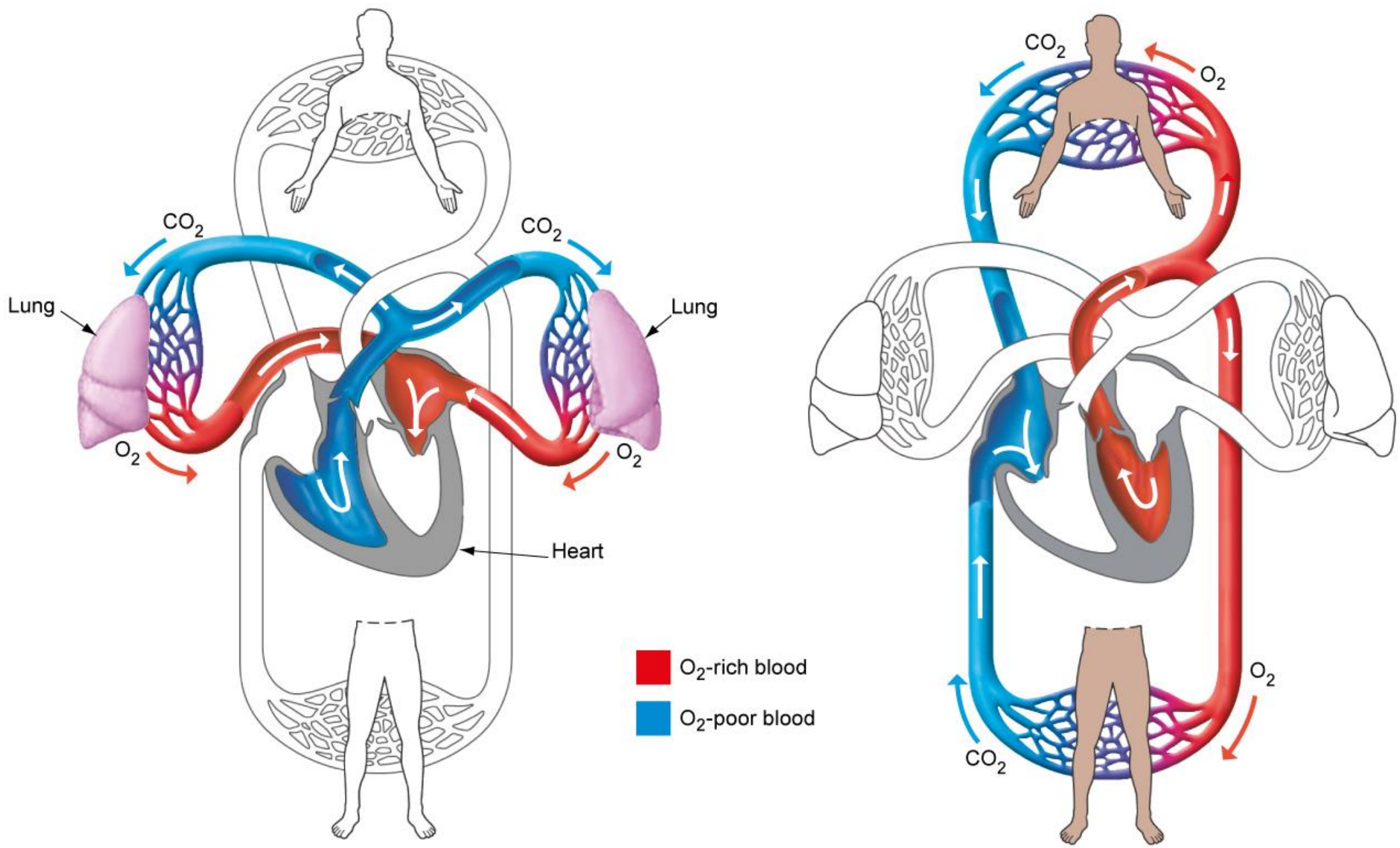
The Diversity of Circulatory Systems (2 of 2)



Unifying Concepts of Animal Circulation

- The **pulmonary circuit** carries blood between the heart and lungs.
  - In the lungs, CO_2 diffuses from the blood into the lungs, while O_2 diffuses from the lungs to the blood.
  - The pulmonary circuit then returns this O_2-rich blood back to the heart.
- The **systemic circuit** carries blood between the heart and the rest of the body.
  - The blood supplies O_2 to body tissues while it picks up CO_2.
  - The O_2-poor blood returns to the heart via the systemic circuit.

The Pulmonary and Systemic Circuits in Humans



(a) **Pulmonary circuit.** In organisms with double circulation, the pulmonary circuit transports blood between the heart and lungs.

(b) **Systemic circuit.** The systemic circuit transports blood between the heart and the rest of the body.

Unifying Concepts of Animal Circulation

- Obstruction of the cardiovascular system is dangerous and sometimes deadly.
  - If a blood clot gets lodged in a lung vessel, it can cause shortness of breath and lung tissue damage.
  - If the clot is large enough, it may completely obstruct blood flow through the pulmonary circuit and cause sudden death as the heart and brain lose access to O_2-rich blood.
  - Clots lodged in the lungs often originate from clots that form in the veins of the legs, a condition known as deep vein thrombosis (DVT). Risk factors for DVT include physical inactivity and dehydration.

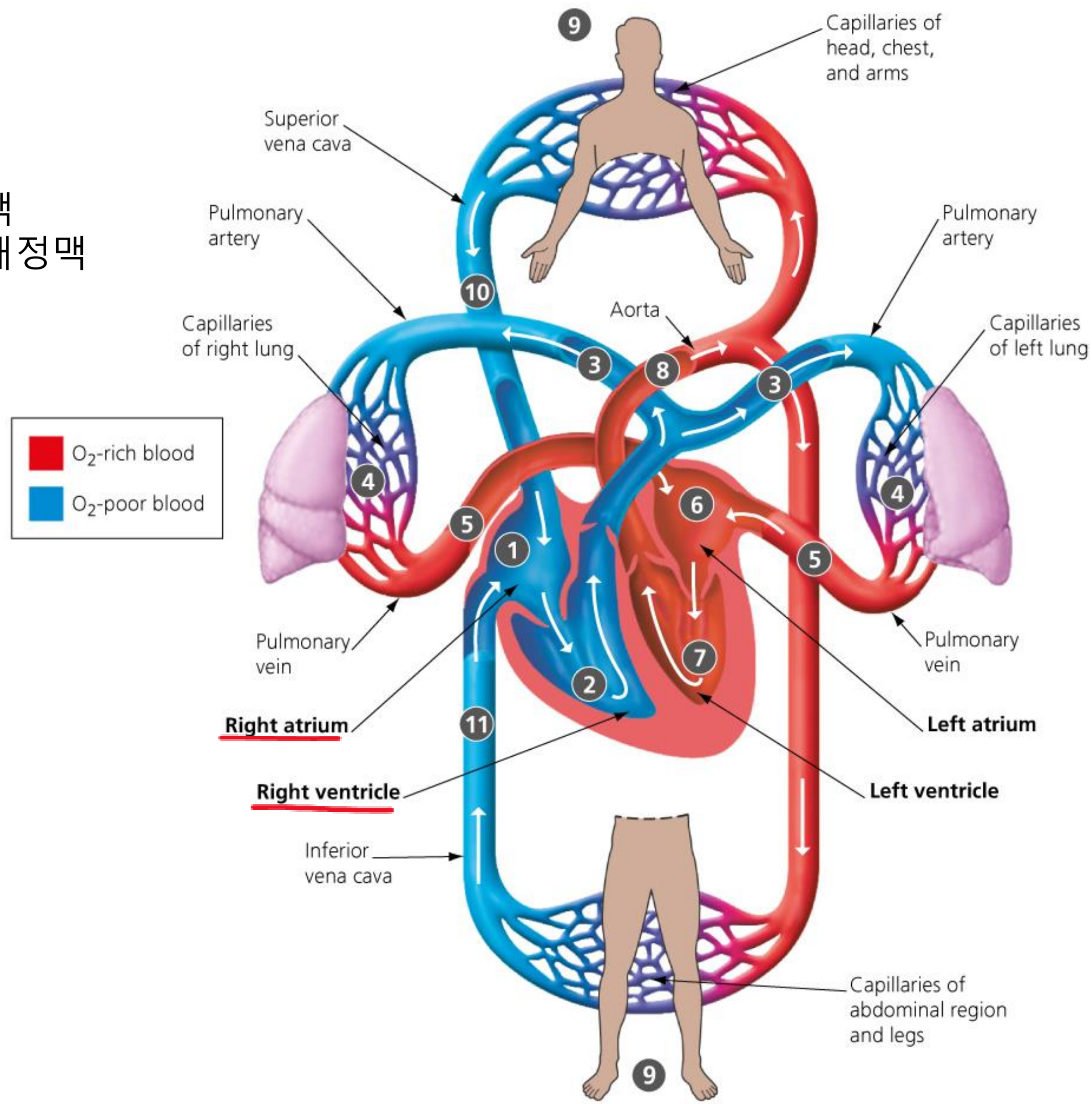
Deep vein thrombosis: 심부 정맥 혈전증

The Human Cardiovascular System: The Path of Blood

- The human cardiovascular system consists of the
  - heart,
  - blood vessels, and
  - circulating blood.
- **Figure 23.3** traces the path of blood as it makes one complete trip around the body.

The Path of Blood Through the Human Cardiovascular

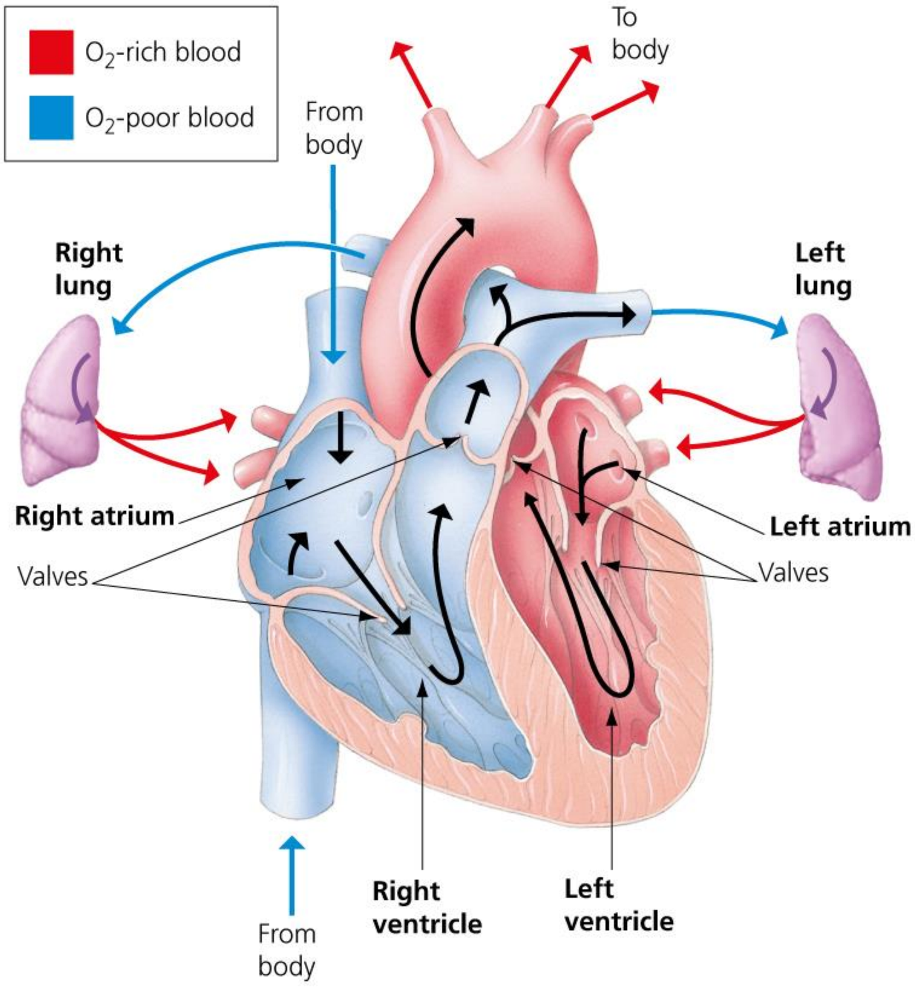
Pulmonary artery: 폐동맥
Superior vena cava: 상대정맥
Artery: 동맥
Atrium: 심방
Ventricle: 심실



How the Heart Works (1 of 2)

- The human **heart** is a muscular organ about the size of a fist, located under the breastbone.
  - Four heart chambers support double circulation and prevent O_2-rich and O_2-poor blood of each circuit from mixing.
  - The blood is forcefully pumped out to body tissues from the left ventricle.

Blood Flow Through the Human Heart



How the Heart Works (2 of 2)

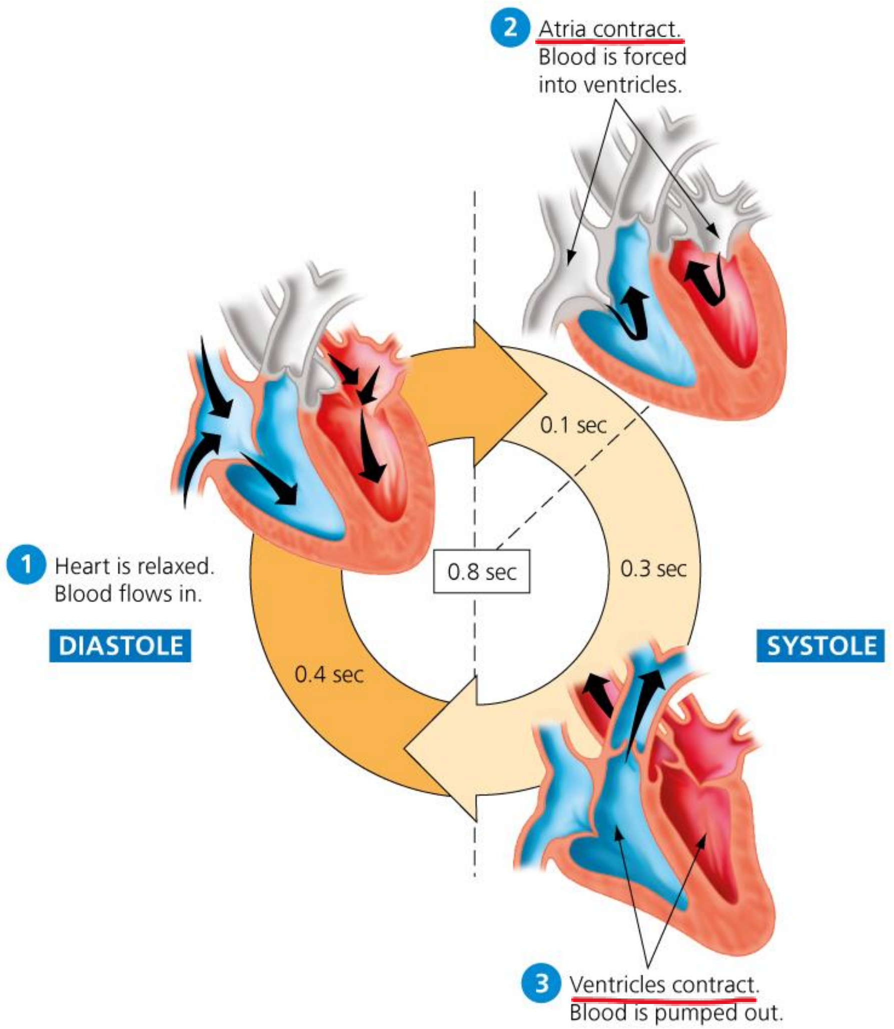
- Amphibians and many reptiles have hearts with only three chambers. The O_2-rich blood mixes with O_2-poor blood in a single ventricle before pumping to the systemic circuit.
- The evolution of a powerful four-chambered heart was an essential adaptation to support the high metabolic needs of birds and mammals, which are endothermic.
  - Endotherms use about ten times as much energy as equal-sized ectotherms.
  - Our four-chambered heart supports our body's need for efficient fuel and O_2 delivery.

The Cardiac Cycle (1 of 2)

- The cardiac muscles of the heart relax and contract rhythmically in the **cardiac cycle**.
  - One heartbeat makes up a complete circuit of the cardiac cycle. A healthy adult at rest has a **heart rate** of about 60 to100 beats per minute.
  - You can measure your heart rate by taking your **pulse** to feel an artery stretch with each heartbeat.
  - The relaxation phase of the heart cycle is known as **diastole**; the contraction phase is called **systole**.
  - **Figure 23.5** follows the heart through one cardiac cycle, which lasts about 0.8 seconds.

Diastole: 확장기
Systole: 수축기

The Cardiac Cycle (2 of 2)

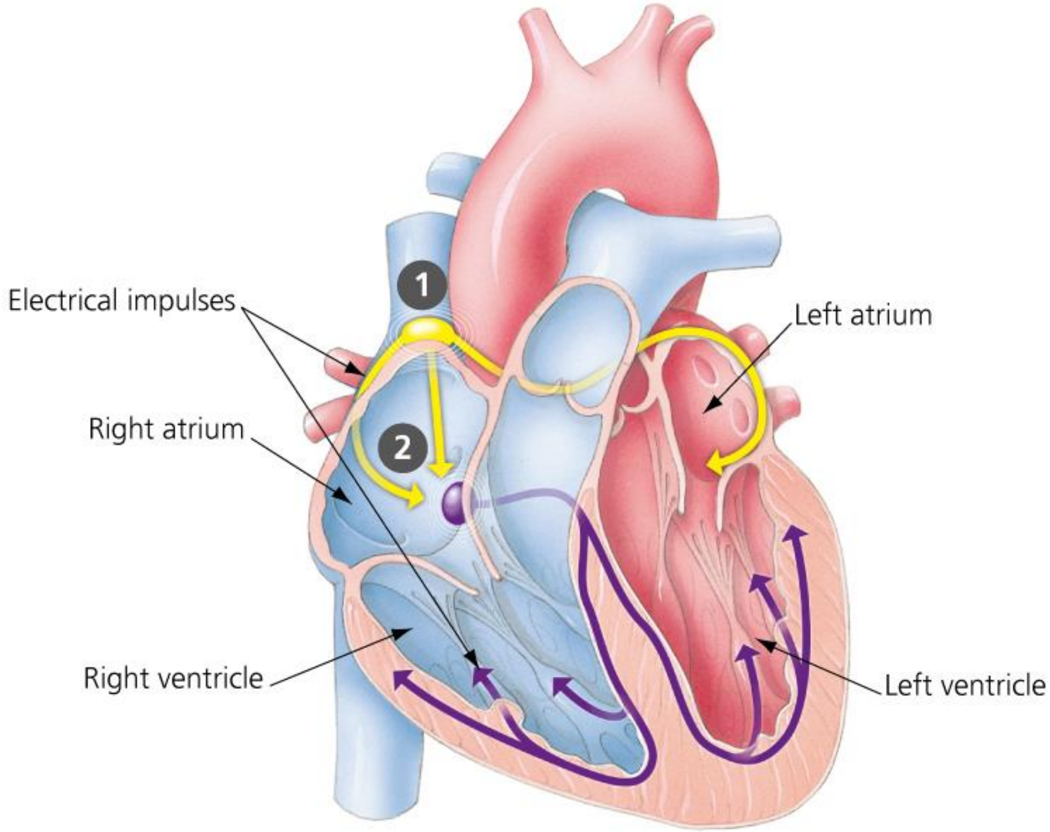


The Pacemaker and the Control of Heart Rate (1 of 2)

- All the muscle cells in your heart beat in unison under the direction of a conductor called the sinoatrial node, or the **pacemaker**.
  - It is made up of specialized muscle tissue in the wall of the right atrium that generates electrical impulses, which spread quickly through the walls of both atria, prompting the atria to contract at the same time.
  - The pacemaker directs muscles of the heart to contract faster or slower under the influence of a variety of signals.
  - **Figure 23.6a** shows the spread of these electrical signals through the heart.

Sinoatrial node: 동방결절
Pacemaker: 심박조율기

The Heart's Natural Pacemaker

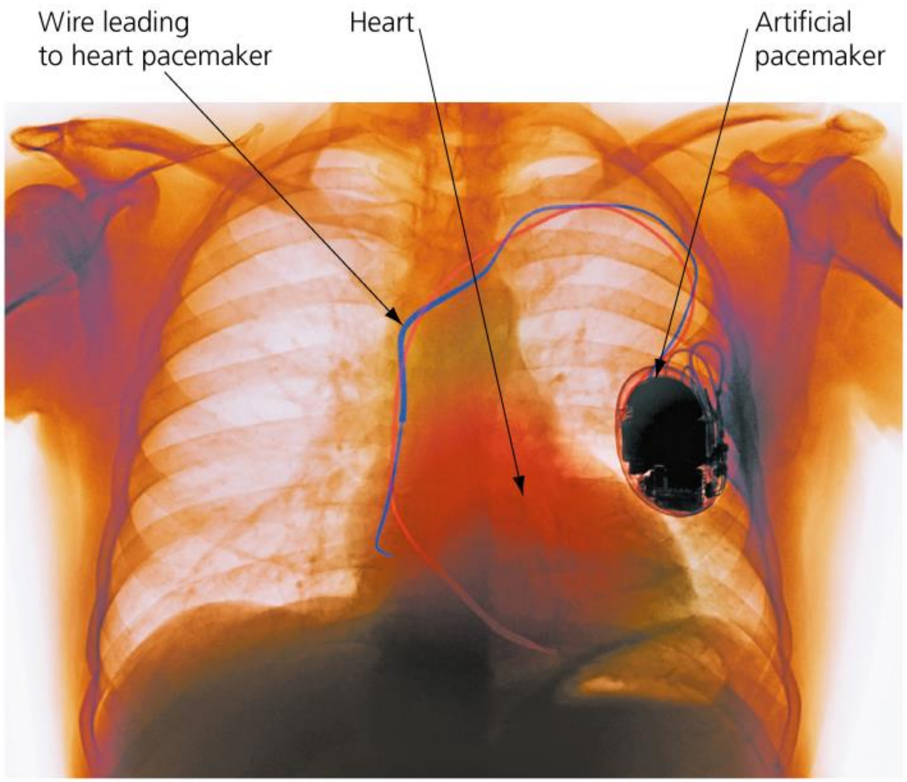


(a) The heart's natural pacemaker. The pacemaker is located in the right atrium. Electrical impulses spread through the heart, first to the atria (shown in yellow arrows), then to the ventricles (purple arrows).

The Pacemaker and the Control of Heart Rate (2 of 2)

- Sometimes the heart's pacemaker fails to coordinate the electrical impulses, and the muscles of the heart contract out of sync, producing an erratic heart rhythm.
- If a heart continually fails to maintain a normal rhythm, an artificial pacemaker that emits rhythmic electrical signals can be surgically implanted into the cardiac muscle to maintain a normal heartbeat.

An Artificial Pacemaker

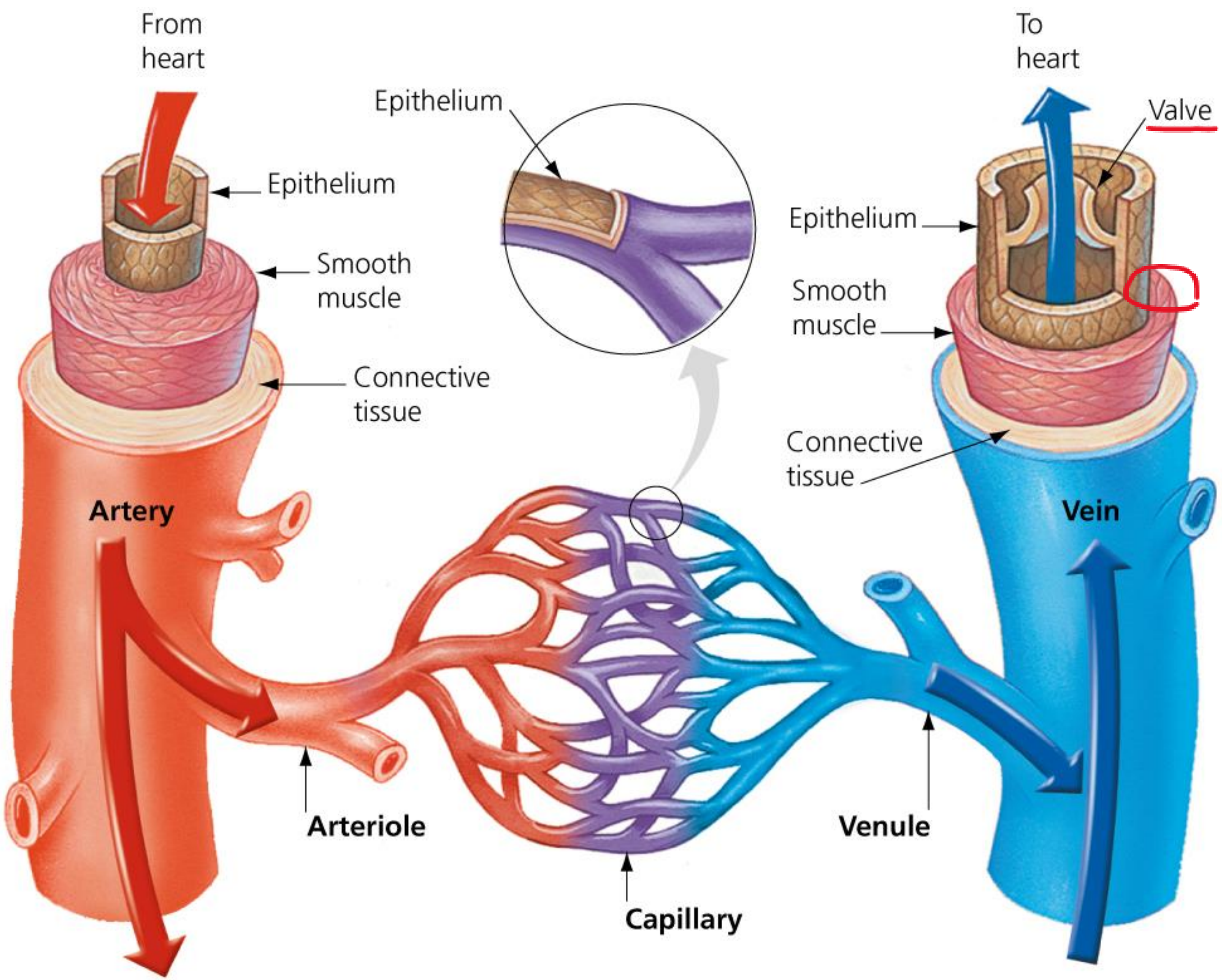


(b) An artificial pacemaker. A small electronic device surgically implanted into cardiac muscle or (as shown here) the chest cavity and connected to the heart's pacemaker by a wire can help maintain proper electrical rhythms in a defective heart.

Blood Vessels

- All blood vessels are lined by a thin layer of tightly packed epithelial cells.
- Structural differences in the walls of the different kinds of blood vessels correlate with their different functions.
- **Figure 23.7** compares the structure of blood vessels.

The Structure of Blood Vessels



Pearson

Blood Flow through Arteries (1 of 2)

- **Blood pressure** is the force that blood exerts against the walls of your arteries.
  - Created by the pumping heart, blood pressure pushes blood from the heart through the arteries, arterioles, and capillary beds.
  - Blood pressure is recorded as two numbers.
    - The first number is blood pressure during systole, when the ventricles contract.
    - The second number is the blood pressure that remains in the arteries during diastole, when the elastic walls of the arteries snap back.

Snap back: 빠르게 회복하다

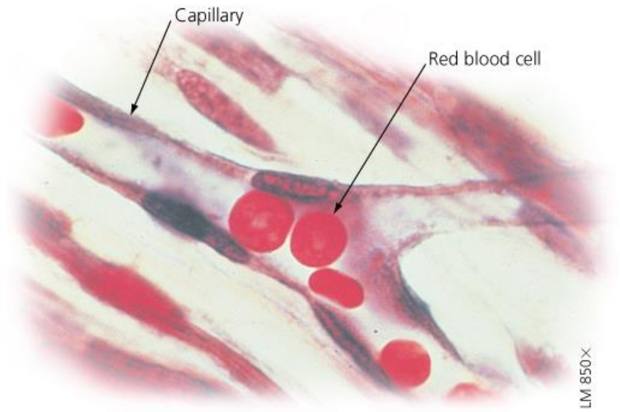
Blood Flow through Arteries (2 of 2)

- Optimal blood pressure for adults is below 120 systolic and below 80 diastolic.
- High blood pressure, or **hypertension**, is persistent systolic blood pressure higher than 140 and/or diastolic blood pressure higher than 90.
- Hypertension is sometimes called a "silent killer" because it often displays no outward symptoms for years while increasing the risks of heart disease, a heart attack, or a stroke.
- To control hypertension, you can eat a healthy diet, avoid smoking and excess alcohol, exercise regularly, and maintain a healthy weight.

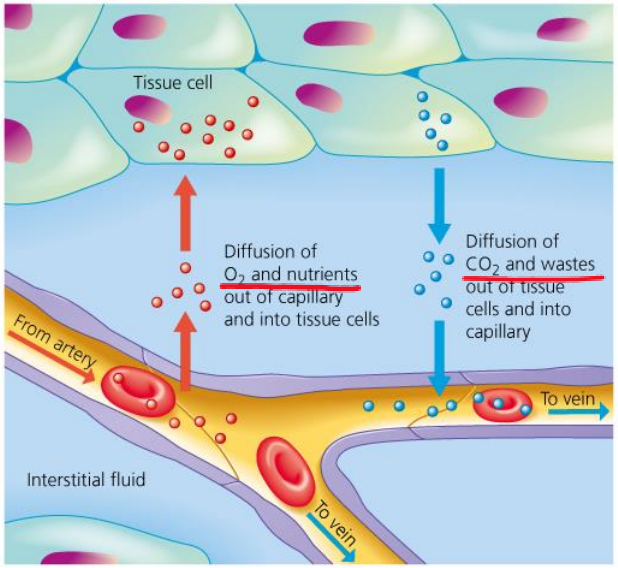
Blood Flow through Capillary Beds

- The most important function of the circulatory system is the chemical exchange between the blood and tissue cells within capillary beds.
  - The walls of capillaries are thin and leaky.
  - Consequently, as blood enters a capillary at the arterial end, blood pressure pushes fluid rich in O_2, nutrients, and other molecules out of the capillary and into the interstitial fluid.
  - Figure 23.8 shows the chemical exchange between the blood and tissue cells.

Chemical Exchange Between the Blood and Tissue Cells



(a) Capillaries. Blood flowing through the circulatory system eventually reaches capillaries, the small vessels where exchange with cells actually takes place.

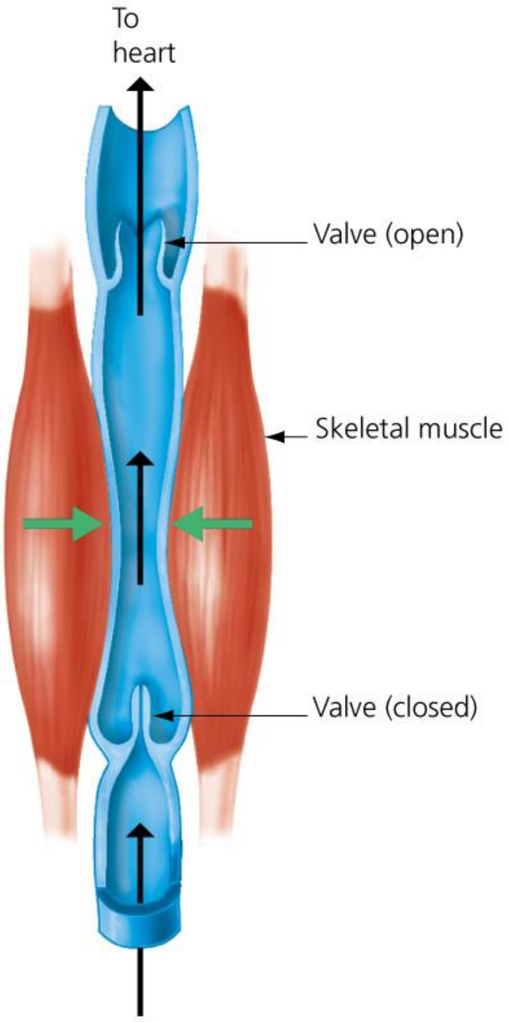


(b) Chemical exchange. Within the capillary beds, there is local exchange of molecules between the blood and interstitial fluid, which bathes the cells of tissues.

Blood Return through Veins (1 of 2)

- After molecules are exchanged between the blood and body cells, blood flows from the capillaries into small venules, then into larger veins, and finally to the inferior and superior venae cavae, the two large blood vessels that flow into the heart.
  - By the time blood enters the veins, the pressure originating from the heart has dropped to near zero.
  - The blood still moves through veins, even against the force of gravity, because veins are sandwiched between skeletal muscles.
  - As these muscles contract (when you walk, for example), they squeeze the blood along.

Blood Flow in a Vein



Blood Return through Veins (2 of 2)

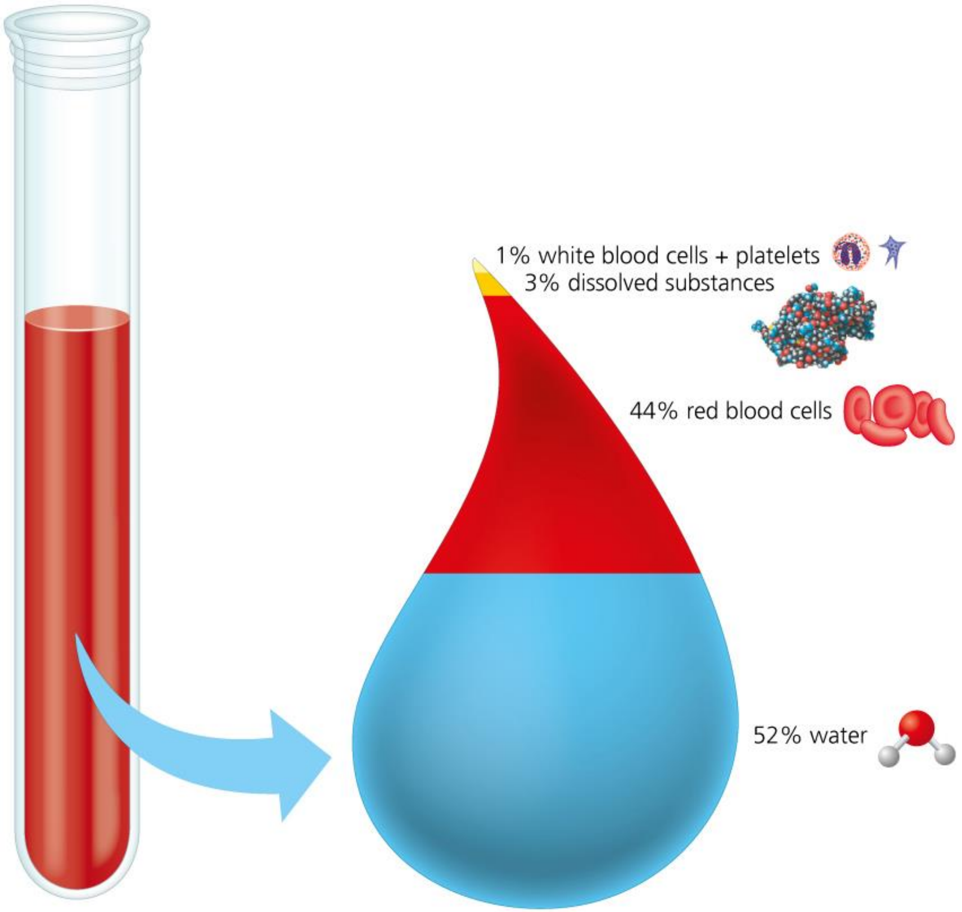
- After a while without contracting muscles, a person will start to become weak and dizzy and could even faint because gravity prevents blood from returning to the heart in sufficient amounts to supply the brain with O_2.
- Over time, leg veins may stretch and enlarge and the valves within them weaken. As a result, veins just under the skin can become visibly swollen, a condition called varicose veins.

Varicose vein: 정맥류성 정맥 (특히 다리의 정맥이 부풀어 아픈 것)

Blood:

- The circulatory system of an adult human has about 5 L of blood.
  - Just over half this volume consists of a yellowish liquid called **plasma**, consisting of water and dissolved salts, proteins, and various other molecules, such as nutrients, wastes, and hormones.
  - Suspended within the plasma are three types of cellular elements: red blood cells, white blood cells, and platelets.

The Composition of Human Blood



Red Blood Cells (1 of 2)

- **Red blood cells**, also called erythrocytes, are by far the most numerous type of blood cell.
  - Each red blood cell contains approximately 250 million molecules of **hemoglobin**, an iron-containing protein that transports oxygen.
  - As red blood cells pass through the capillary beds of your lungs, O_2 diffuses into them and binds to the hemoglobin.
  - This process is reversed in the capillaries of the systemic circuit, where the hemoglobin unloads its cargo of O_2 to the body's cells.

Red Blood Cells (2 of 2)

- The structure of a red blood cell supports its function. Human red blood cells
  - are shaped like disks with indentations in the middle, increasing the surface area available for gas exchange, and
  - lack nuclei and other organelles, leaving more room to carry hemoglobin.
- **Anemia** occurs when there is a low amount of hemoglobin or a low number of red blood cells.
- Erythropoietin (EPO) is a hormone made by the kidneys. It stimulates the bone marrow's production of O_2-carrying red blood cells.

Blood Doping

- Athletes sometimes abuse synthetic EPO to enhance their blood oxygen level, a practice referred to as blood doping.
  - Blood doping is difficult to detect because EPO is a hormone produced naturally by the body and because synthetic EPO is rapidly cleared from the bloodstream.
  - One way that athletic commissions test for EPO abuse is by measuring the percentage of red blood cells in blood volume over time.
  - High percentages and erratic spikes in these values are grounds for disqualification.

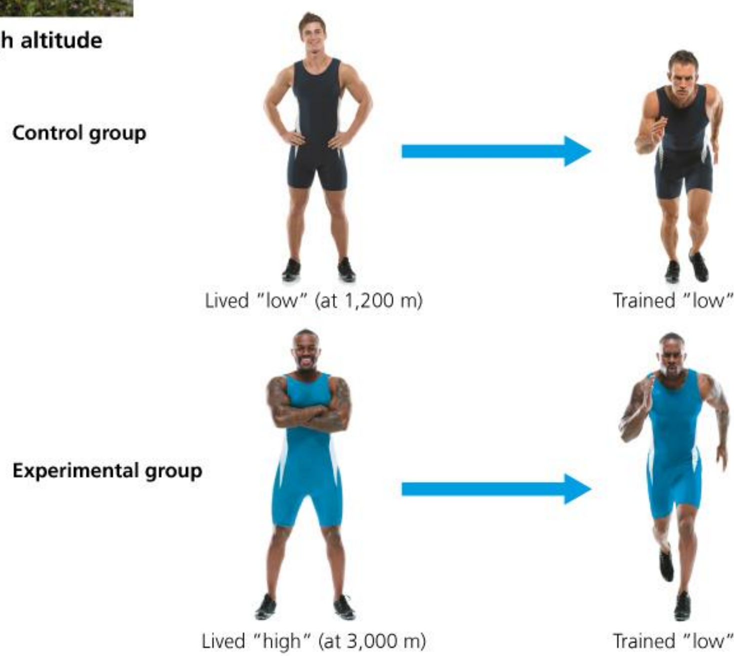
The Process of Science: Live High, Train Low? (1 of 2)

- **Background**: If an athlete stays at high altitude for weeks or months, the kidneys will compensate for the low O_2 level in the body by producing more EPO, which results in more red blood cells, which in turn results in more O_2 carried to muscle tissues. Will living at high altitudes boost red blood cell production, which can then improve stamina when training and competing at lower altitudes?

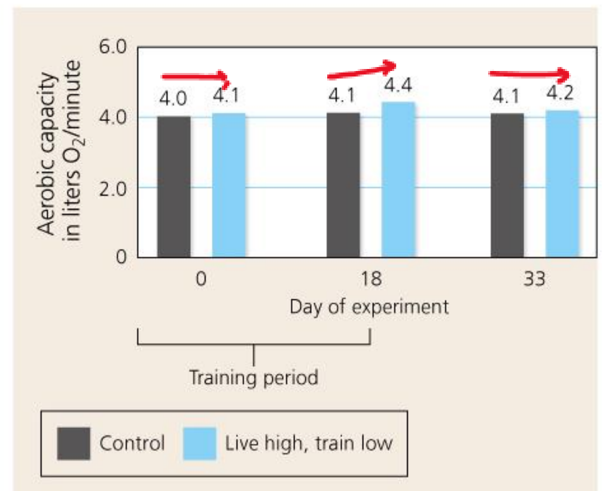
- **Method**: Researchers recruited 11 elite runners. Five lived "high" for 18 days in rooms adjusted to simulate an altitude of 3,000 m. The other six lived "low" at 1,200 m. All athletes trained at 1,200 m.

Live High, Train Low

(a) **Athletic competitions at high altitude require greater endurance.**



(b) **The experimental protocol for the study**



(c) **The results of this study suggest that living at high altitude while training at low altitude can improve athletic performance, but the effect fades.**

The Process of Science: Live High, Train Low? (2 of 2)

- **Results**:
  - At the start of the study, both groups were similar in their aerobic capacity, a measure of the body's ability to take in and use O_2.
  - By the end of the training regimen 18 days later, the "live high, train low" group had a higher aerobic capacity that gradually fell over two weeks.
  - Although these results are clear, there were too few test subjects to draw definitive conclusions. This is a common problem when using human test subjects, especially ones as rare as endurance athletes.

White Blood Cells and Defense

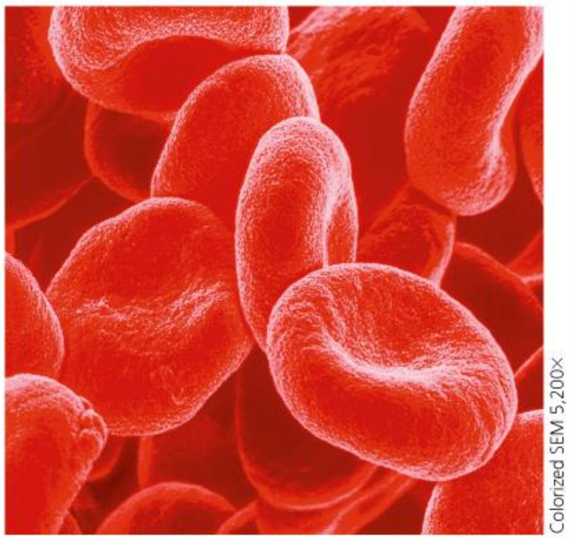
- **White blood cells** (leukocytes) contain nuclei and other organelles, are larger and lack hemoglobin, are less abundant than red blood cells, but temporarily increase in number when the body is combating an infection.

Pearson

The Three Cellular Components of Blood

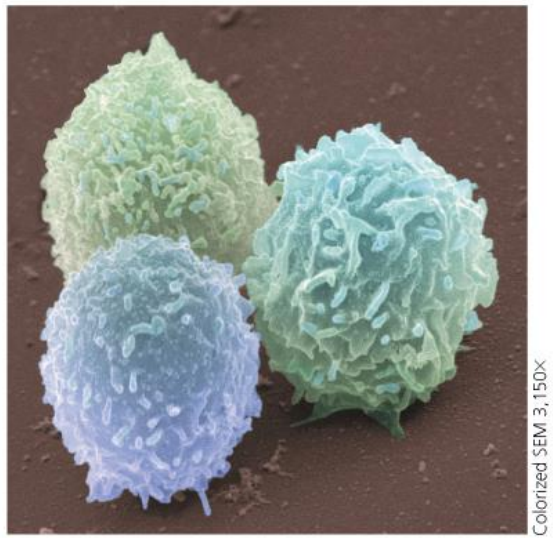
CELLULAR COMPONENTS OF BLOOD

Red Blood Cells
(cells that carry oxygen)



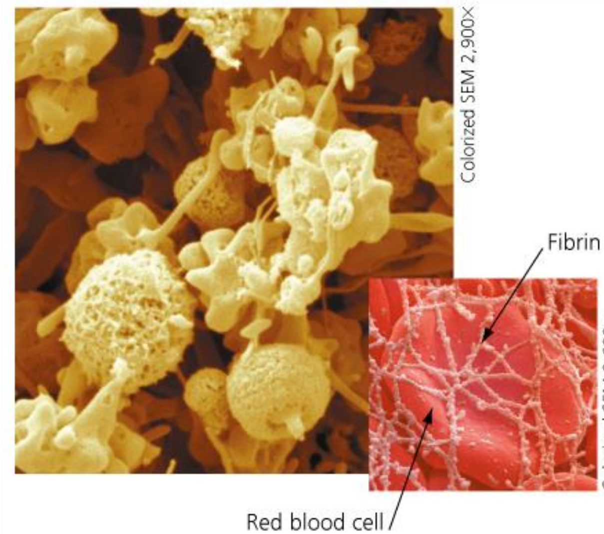
This electron micrograph shows the indented disk shape of human red blood cells.

White Blood Cells
(cells that fight infection)



This electron micrograph shows lymphocytes, a type of white blood cell.

Platelets
(bits of membrane-enclosed cytoplasm that aid clotting)



Platelets help produce a clotting protein called fibrin (shown at bottom wrapping around a red blood cell).

Platelets and Blood Clotting

- Blood contains two components that help form clots: platelets and fibrinogen.
    1. **Platelets** are bits of membrane pinched off from larger cells in the bone marrow (megakaryocytes).
    2. Platelets also release clotting factors, molecules that convert **fibrinogen**, a protein found in the plasma, to a threadlike protein called **fibrin**. Molecules of fibrin form a dense network to create a patch.
    - In the disease hemophilia, a genetic mutation in a gene for a clotting factor results in excessive, sometimes fatal bleeding.

Donating Blood

- Excessive bleeding and certain diseases can be treated through blood transfusion, the process by which blood is intravenously (through veins) transferred from one person into another.
  - Transfusion is made by blood donation.
  - When whole blood is donated, all the components (plasma, red blood cells, white blood cells, and platelets) are collected and then separated.
  - Partial blood donation is possible. In apheresis, only platelets, plasma, or stem cells are removed from a donor’s blood.

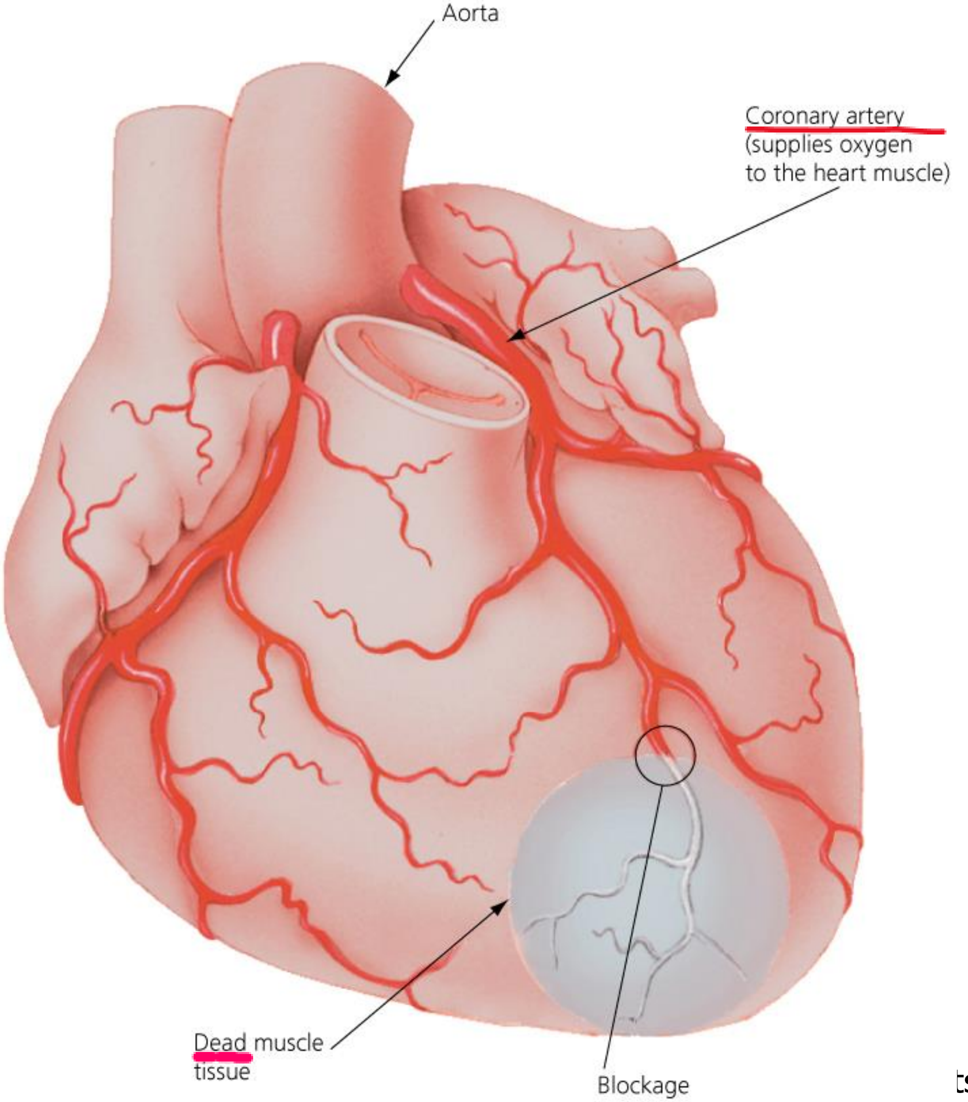
Apheresis: 성분채집술, 성분 헌혈

Cardiovascular Disease

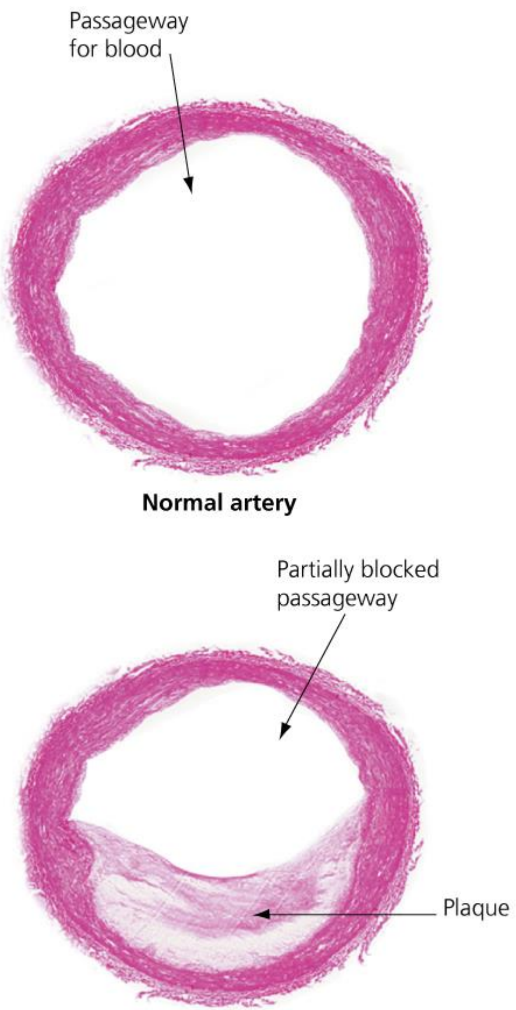
- Diseases of the heart and blood vessels are collectively called **cardiovascular disease**, and they account for one in three deaths in the United States. **Coronary arteries** are the vessels that supply O_2-rich blood to the heart muscle.
  - Chronic disease of the coronary and other arteries throughout the body is called **atherosclerosis**, in which plaque develops in the inner walls of arteries.
  - A coronary artery partially blocked by plaque may cause angina, occasional chest pain.
  - A fully blocked coronary artery will cause quick death of heart muscle cells.

Coronary artery: 관상동맥
Atherosclerosis: 동맥경화증
Angina: 협심증

Blockage of a Coronary Artery, Resulting in a Heart Attack



A Normal Artery and an Artery Showing Atherosclerosis



Normal artery

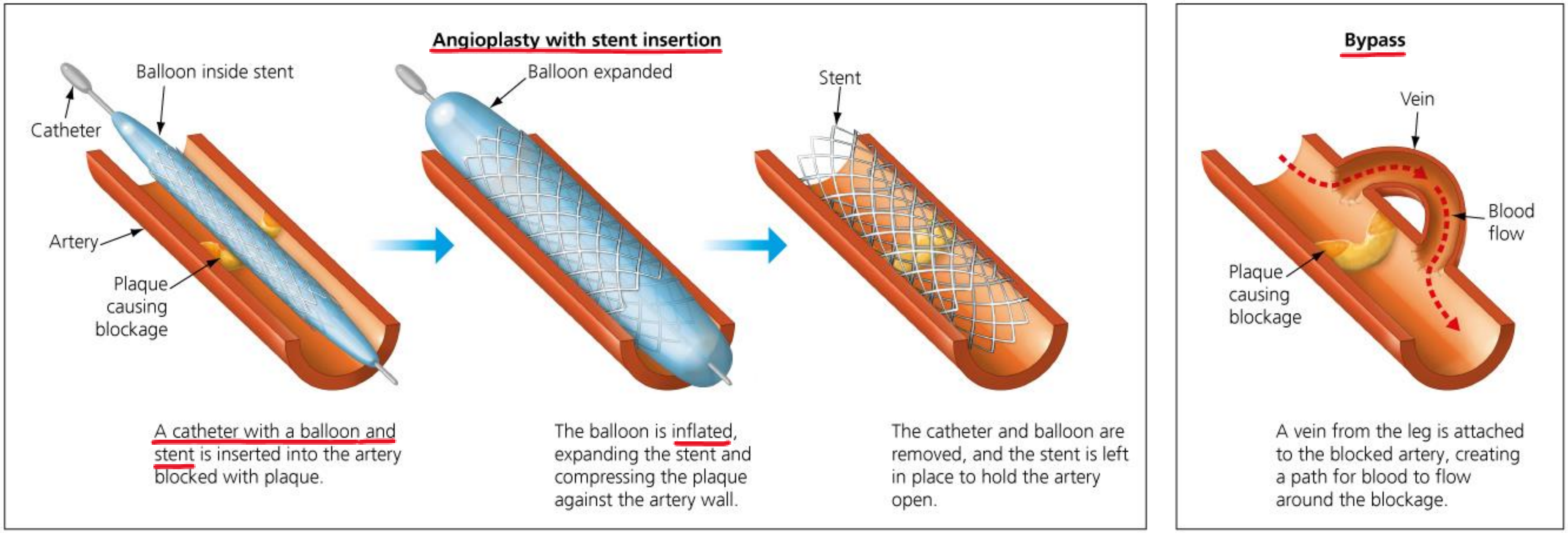
Artery partially blocked by plaque

Cardiovascular Disease (2 of 3)

- Approximately one-third of heart attack victims die almost immediately. For those who survive, the ability of the damaged heart to pump blood may be seriously impaired for life because heart muscle cannot be replaced or repaired.
  - Certain drugs can lower the risk of developing clots.
  - Angioplasty is the insertion of a tiny catheter with a balloon that is inflated to compress the plaque and widen clogged arteries. A small wire mesh stent, to open an artery, may be inserted during the angioplasty process.
  - Bypass surgery is a much more drastic remedy.

Angioplasty: 혈관형성술

Methods of Repairing Arteries Blocked by Plaque



Pearson

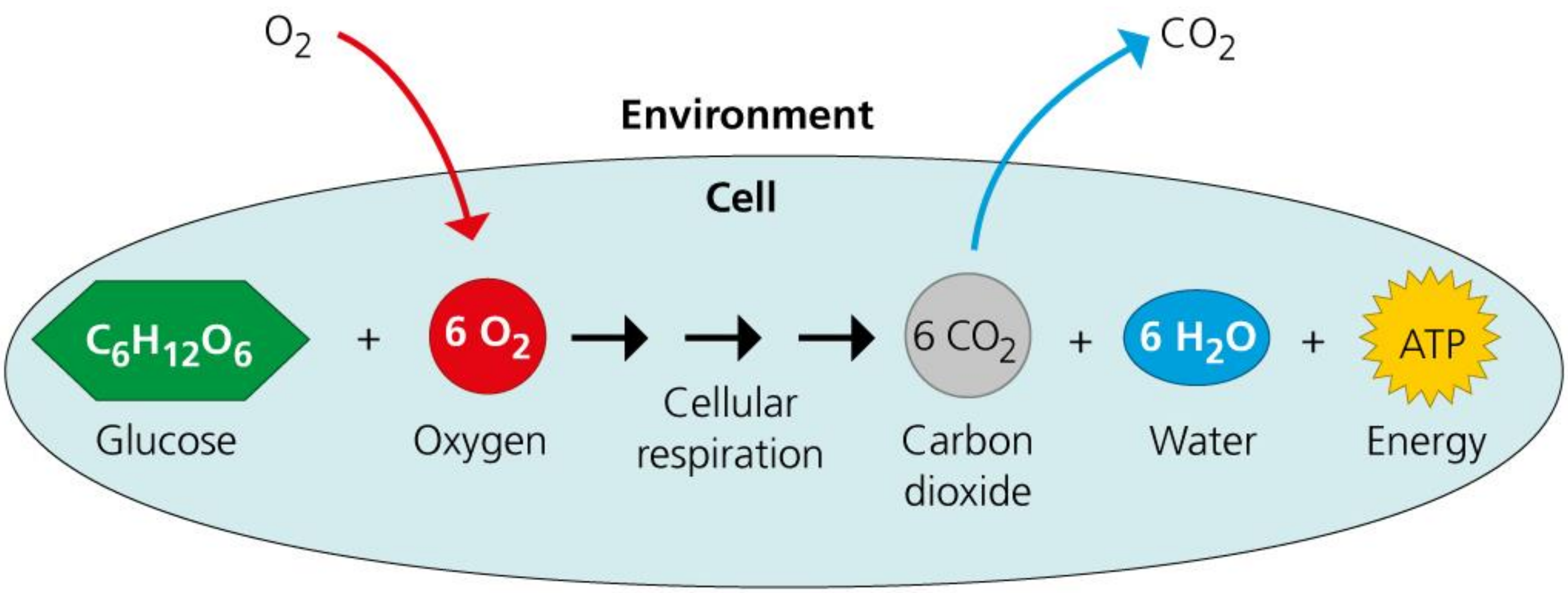
Cardiovascular Disease (3 of 3)

- Cardiovascular disease involves inherited factors but can be reduced by
  - not smoking,
  - exercising regularly, and
  - eating a healthy diet high in fruits, vegetables, and whole grains.
- The death rate from cardiovascular disease in the United States has been cut in half during the past 50 years.

Unifying Concepts of Animal Respiration
(1 of 6)

- Cellular respiration uses oxygen and glucose and produces water, carbon dioxide, and energy in the form of ATP.
  - All working cells therefore require a steady supply of oxygen from the environment and must continuously dispose of carbon dioxide.

Unifying Concepts of Animal Respiration
(2 of 6)



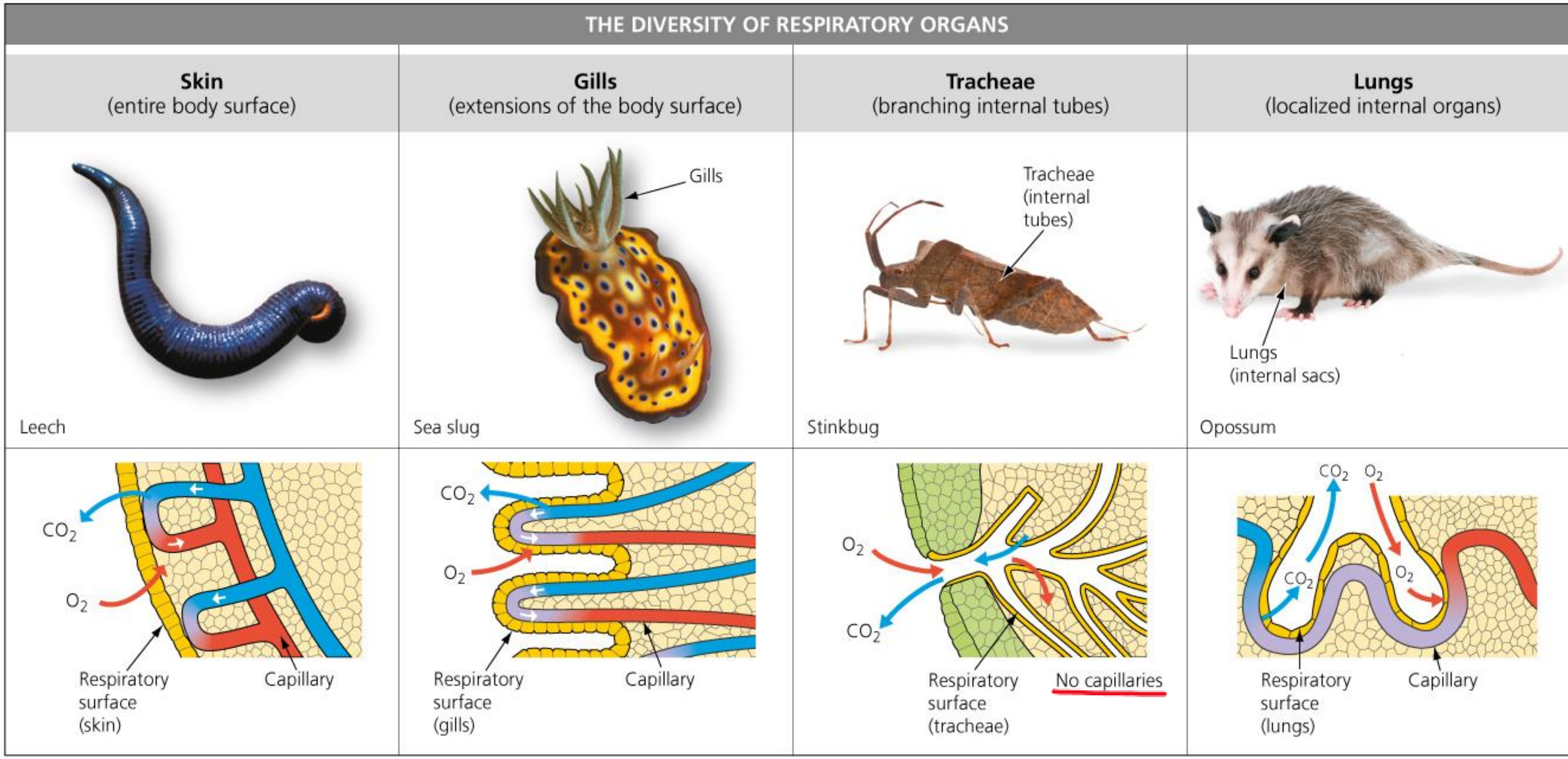
Unifying Concepts of Animal Respiration

- The **respiratory system** consists of several organs that facilitate exchange of O_2 and CO_2 between the environment and cells.
  - The interaction of bodily systems (the circulatory and respiratory systems) allows for life's processes (cellular respiration) to occur.
  - The part of an animal where O_2 from the environment diffuses into living cells and CO_2 diffuses out to the surrounding environment is the respiratory surface. It is usually thin, moist, and covered with a single layer of cells, characteristics that allow rapid diffusion between the body and the environment.

Unifying Concepts of Animal Respiration

- Within the animal kingdom, a variety of respiratory surfaces have evolved.
  - For animals such as sponges and flatworms, the plasma membrane of every cell in the body is close enough to the outside environment for gases to diffuse in and out.
  - In many animals, however, the bulk of the body does not have direct access to the outer environment, and the respiratory surface is the place where the environment and the blood meet.
  - Animals such as leeches, earthworms, and frogs, use their entire outer skin as a respiratory surface.

The Diversity of Respiratory Organs



Unifying Concepts of Animal Respiration
(5 of 6)

- For most animals, the outer surface is either impermeable to gases or lacks sufficient surface area to exchange gases for the whole body.
  - In such animals, specialized regions of the body surface have extensively folded or branched tissues that provide a large surface for gas exchange.
  - **Gills** are outfoldings of the body surface that are suspended in water and found in most aquatic animals. As water passes over the large respiratory surface of the gills, gases diffuse between the water and the blood.

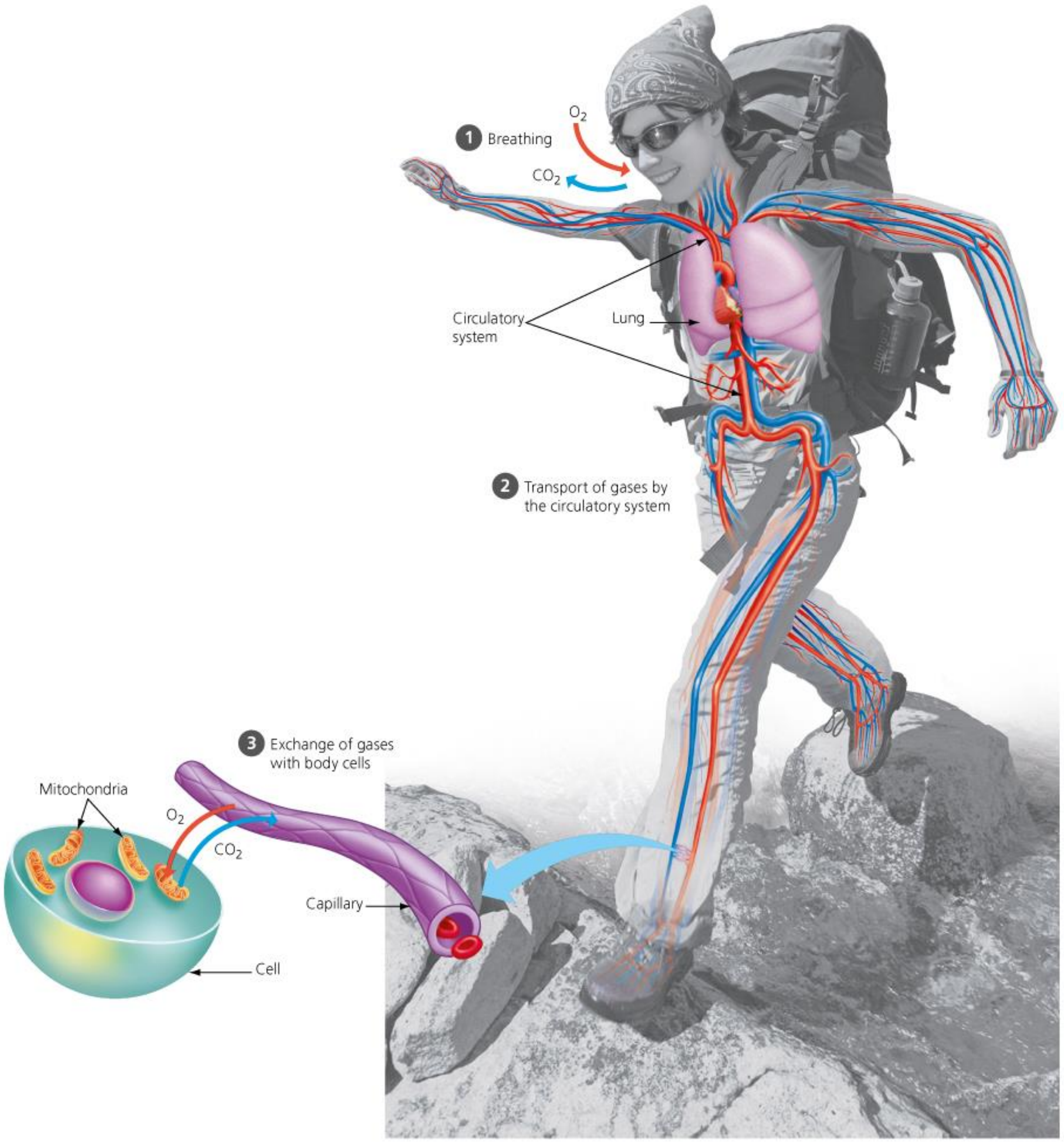
Unifying Concepts of Animal Respiration

- In most land-dwelling animals, the respiratory surfaces are folded into the body and open to the air only through narrow tubes. The two types of respiratory structures seen in land-dwelling animals are tracheal systems and lungs.
  - Insects use a **tracheal system**, an extensive network of branching internal tubes called tracheae. Tracheae begin near the body's surface and branch to narrower tubes that extend to nearly every cell.
  - **Lungs** are localized organs lined with moist epithelium. Gases are carried between the lungs and the body cells by the circulatory system.

The Human Respiratory System (1 of 2)

- The human respiratory system has three phases of gas exchange:
  1. **breathing**, the ventilation of the lungs by alternate inhalation and exhalation,
  2. transport of O_2 from the lungs to the rest of the body via the circulatory system, and
  3. diffusion of O_2 from red blood cells into body cells. The delivered O_2 is used by the body cells to make ATP from food via the process of cellular respiration. This same process produces CO_2 as a waste product that diffuses from cells to the blood.

Gas Exchange



Pearson

The Path of Air (1 of 3)

- Let's follow the flow of air into the lungs.
  - Air enters the respiratory system through the nostrils and mouth. In the nasal cavity, the air is filtered by hairs and mucus, warmed, humidified, and sampled by smell receptors.
  - The air passes to the **pharynx**, where the digestive and respiratory systems meet.
  - From the pharynx, air is inhaled into the **larynx** (voice box) and then into the **trachea** (windpipe).
  - The trachea forks into two **bronchi** (singular, *bronchus*), one leading to each lung.

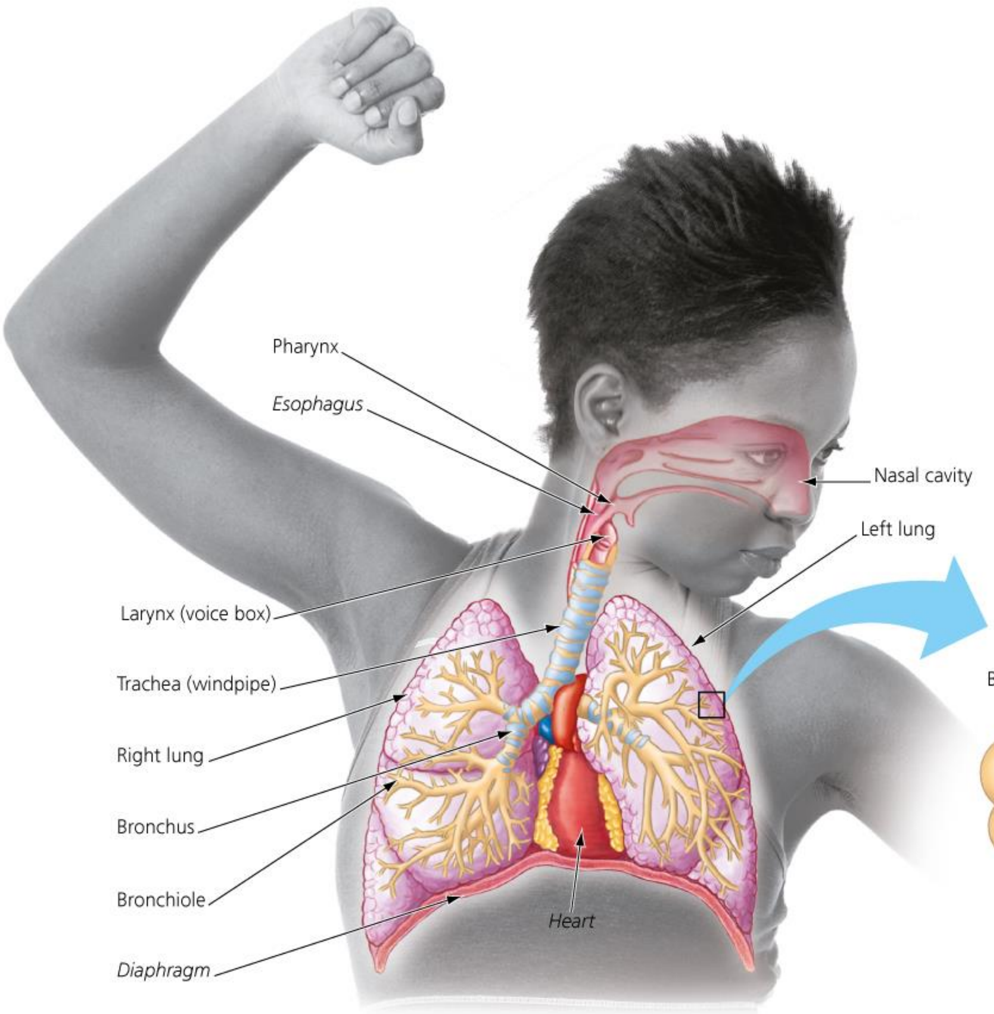
Larynx: 후두
Bronchus: 기관지

The Path of Air (2 of 3)

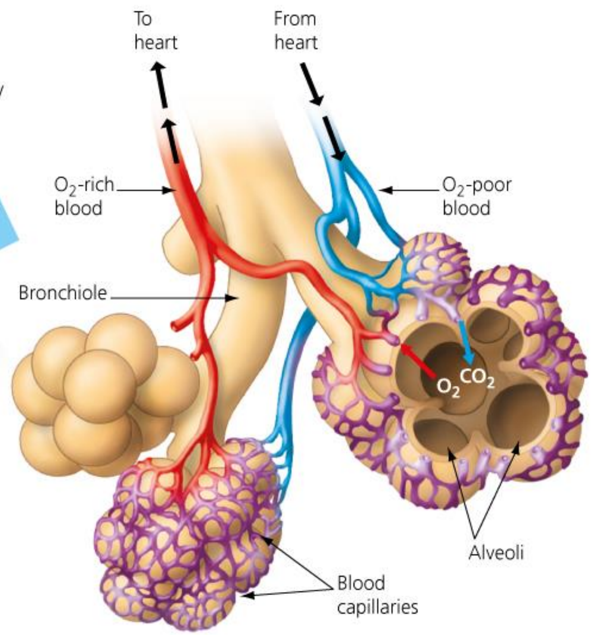
- Within the lungs, each bronchus branches repeatedly into finer and finer tubes called bronchioles.
  - The bronchioles dead-end in grapelike clusters of air sacs called **alveoli** (singular, *alveolus*).
    - Each of your lungs contains millions of these tiny sacs that provide about 50 times more surface area than your skin.
    - The inner surface of each alveolus is lined with a layer of epithelial cells, where the exchange of gases actually takes place.

Bronchioles: 세기관지
Alveoli: 허파꽈리, 폐포

The Human Respiratory System (2 of 2)



(a) Overview of the human respiratory system.



(b) The structure of alveoli. Capillaries surround air sacs called alveoli. These are the sites of gas exchange in the lungs.

The Path of Air (3 of 3)

- The tiny alveoli are delicate and easily damaged, and after age 20 they are not replaced. Destruction of alveoli (usually by smoking, but also by involuntary exposure to air pollution) causes the lung disease emphysema.
- As the air we have been following reaches its final destination, O_2 enters the bloodstream by diffusing from the air into a web of blood capillaries that surrounds each alveolus.
- Your exhale reverses the process: CO_2 diffuses from blood in your capillaries into the alveoli and then moves through your bronchioles, bronchus, trachea, and out of your body.

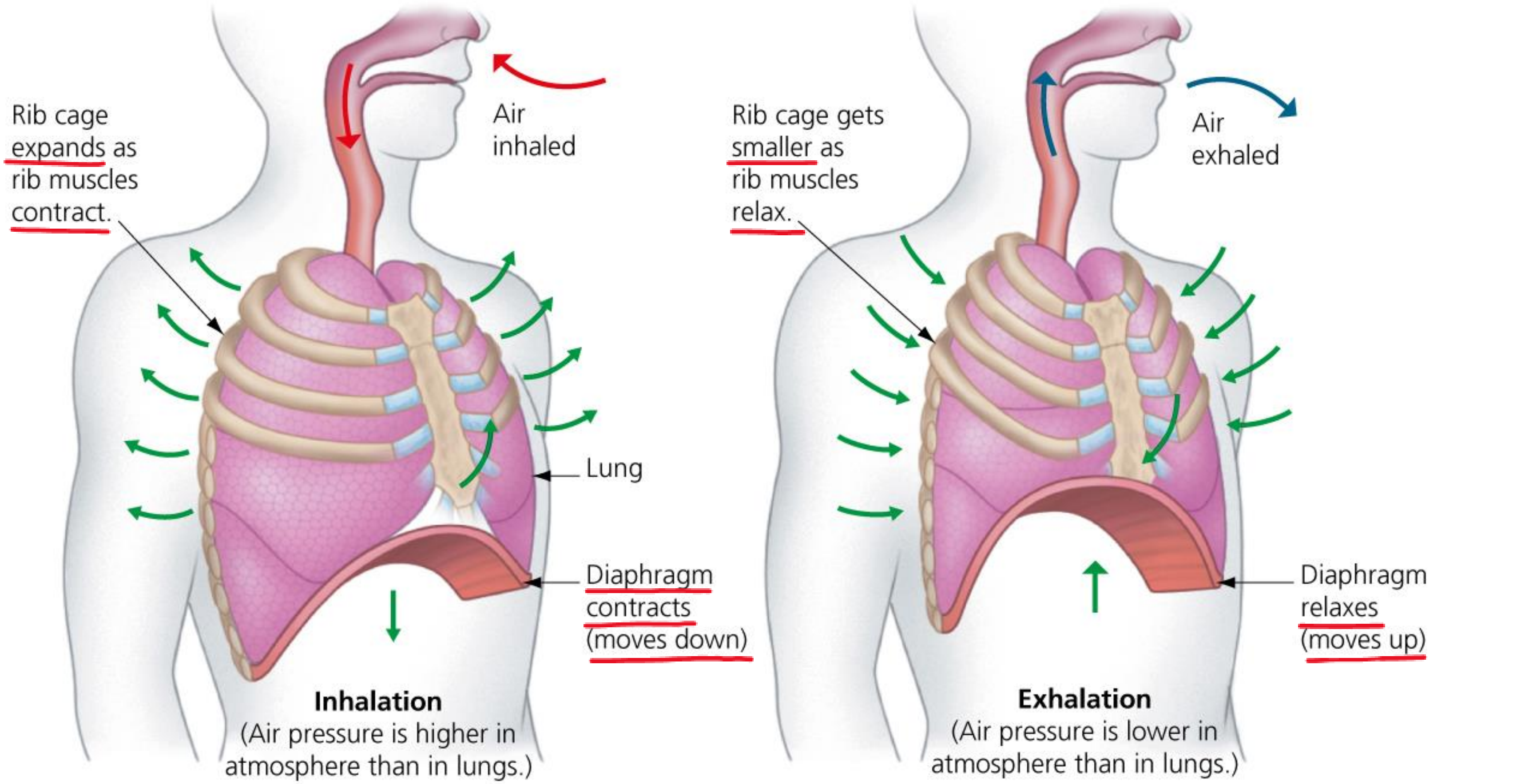
Emphysema: (폐)기종

The Brain's Control over Breathing (1 of 2)

- Breathing is the alternating process of inhalation and exhalation.
  - During inhalation, the chest is expanded by the upward movement of the ribs and downward movement of the **diaphragm**, a sheet of muscle.
  - Air moves into the lungs by negative pressure breathing, as the air pressure in the lungs is lowered by the expansion of the chest cavity.

Diaphragm: 횡경막

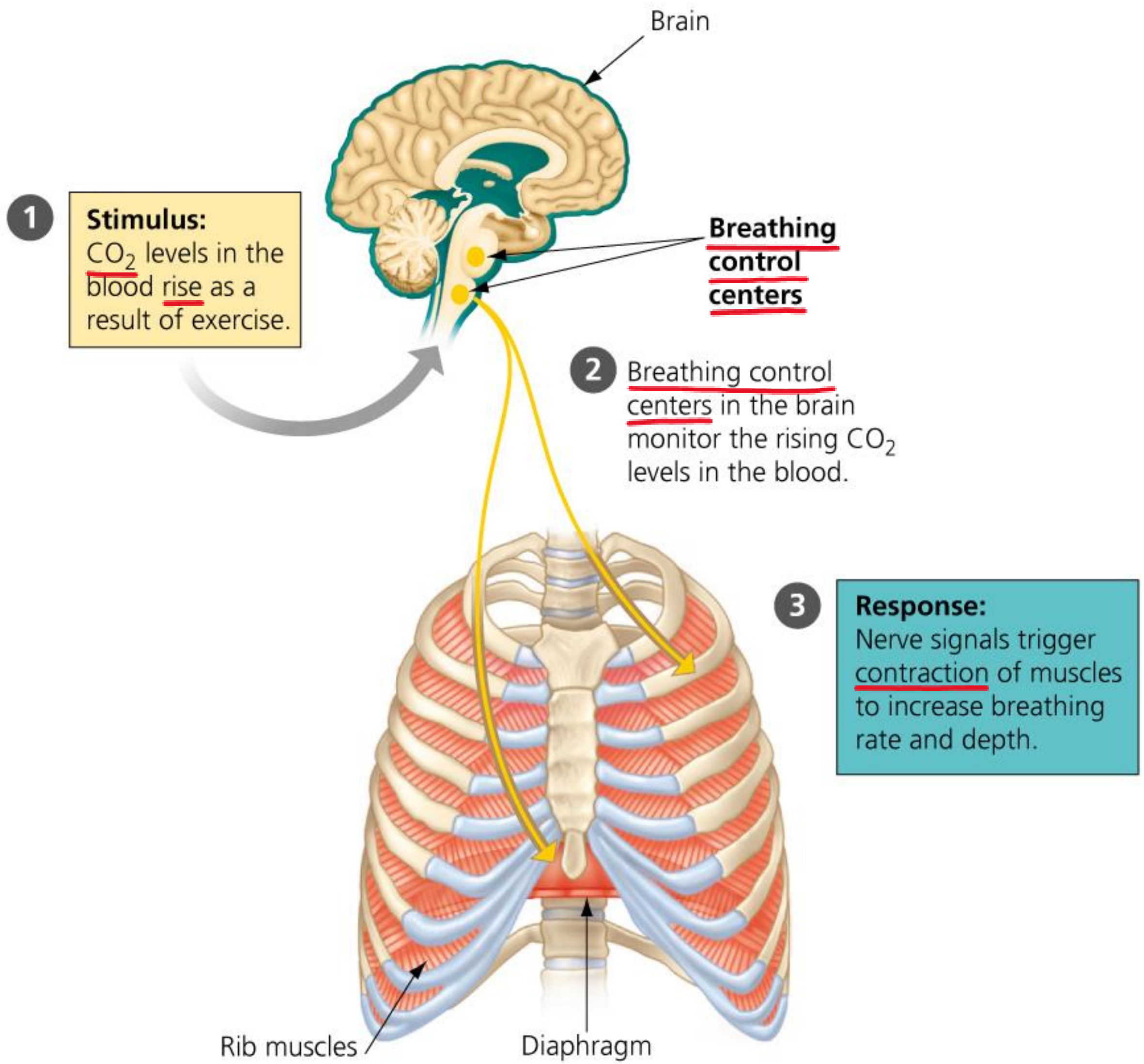
How a Human Breathes



The Brain's Control over Breathing (2 of 2)

- During exhalation, the rib and diaphragm muscles relax, decreasing the volume of the chest cavity. This decreased volume increases the air pressure inside the lungs, forcing air to rush out of the respiratory system.
- You can consciously speed up or slow down your breathing. Most of the time, nerves from breathing control centers in the brainstem maintain a respiratory rate of 10–14 inhalations per minute.
  - This rate can vary, however, like when you exercise.
  - Figure 23.20 highlights this respiratory control system.

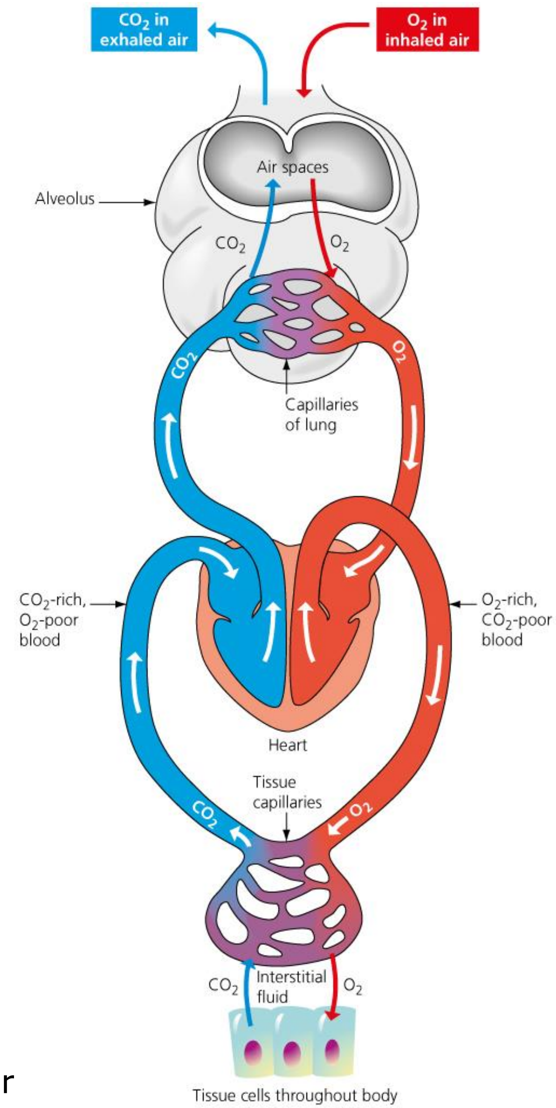
Breathing Control Centers



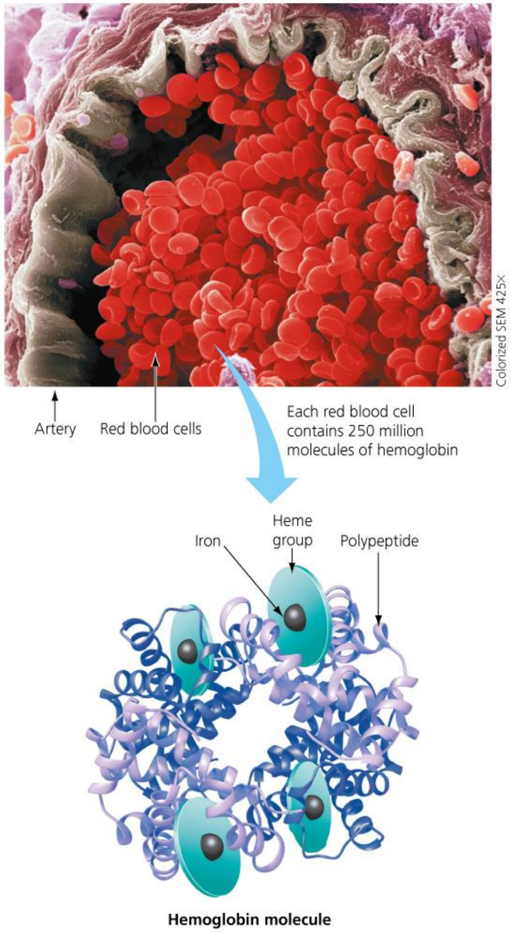
The Role of Hemoglobin in Gas Transport
(1 of 2)

- The human respiratory system takes O_2 into the body and expels CO_2, but it relies on the circulatory system to shuttle these gases between the lungs and the body's cells. But there is a problem with this simple scheme.
  - O_2 does not readily dissolve in blood, so O_2 does not tend to move from the air into the blood on its own. The O_2 binds to hemoglobin, which consists of four polypeptide chains.
  - Hemoglobin loads up on O_2 in the lungs, transports it through the blood, and unloads it at the body's cells.

Gas Transport and Exchange in the Body



A Molecule of Hemoglobin



The Role of Hemoglobin in Gas Transport
(2 of 2)

- A shortage of iron causes less hemoglobin to be produced by the body. Iron deficiency is the most common cause of anemia.
- Carbon monoxide (CO) is a colorless, odorless, poisonous gas that can bind to hemoglobin, even more tightly than O_2 does.
  - Breathing CO can therefore interfere with the delivery of O_2 to body cells, blocking cellular respiration, and causing rapid death.
  - Millions of Americans willingly inhale CO in the form of cigarette smoke.

The Toll of Smoking on the Lungs (1 of 2)

- Every breath you take exposes your respiratory tissues to potentially damaging chemicals.
  - One of the worst sources of air pollution is cigarette smoke. More than 4,000 different chemicals are contained in cigarette smoke, many of which are known to be toxic and even potentially deadly.
  - Smoking slowly damages the respiratory system and leads to chronic obstructive pulmonary disease (COPD), characterized by a chronic cough, difficulty in breathing, irritated epithelial tissue lining the bronchioles, and damaged alveoli which lose their elasticity, decreasing their ability to expel air and contribute to gas exchange.

COPD: 만성폐쇄성폐질환

The Toll of Smoking on the Lungs (2 of 2)

- Lung cancer kills more Americans than any other form of cancer by a wide margin.
  - A teenager who starts smoking shortens his or her life span by over 10 years.
  - Smoking and secondary exposure are still responsible for about one in five deaths every year in the United States, more than all the deaths caused by accidents, alcohol and drug abuse, HIV, and murders combined.
  - No lifestyle choice can have a more positive impact on your long-term health than not smoking.

Healthy versus Cancerous Lungs

(a) Healthy lung (nonsmoker)

(b) Cancerous lung (smoker)

Evolution Connection: Evolving Endurance

- Conditioning can boost athletic endurance by improving the ability of the circulatory and respiratory systems to deliver oxygen to muscles.
- Many Tibetans live and work at altitudes above 13,000 feet and have evolved the ability to thrive at high altitude.
  - They have a higher frequency of versions of genes that contribute to the functioning of the circulatory and respiratory systems.
  - These genes are otherwise rare in low-dwelling Chinese groups.

A Tibetan Mountaineer

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